

Sanjog Thapa

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SUMMARY

Software Engineer with 5+ years of experience. Contributed to building and improving software solutions across various domains including Earth Data Science, Health, and Digital Ads domains with major focus on reliability, scalability, and standard development practices.

WORK EXPERIENCE

NASA IMPACT

Huntsville, AL, USA

Software Engineer II

Mar 2022 - Current

- Developed and managed test-driven deployments at US Greenhouse Gas Center (earth.gov/ghgcenter), performing biweekly upgrades of front-end and back-end services, ensuring 99% uptime through regular maintenance, bug fixes, and quality assurance testing. Technologies used: **Python, ReactJS, AWS (CDK, S3, ECR, Cloudformation, Lambda), Terraform, GitHub Actions**
- Improved the response time of **Features API** by 65% using query execution optimization and cache, hence enabling swift data ingest and visualization of the Greenhouse gases collected by tower stations as a part of inter-agency efforts from NASA, NOAA, and NIST.
- Deployed and managed **Fast API** based Spatial Asset Catalog API for the GIS vector and raster datasets, improving dataset search and discovery for various science use cases
- Implemented data transformation workflow using **Airflow DAGs** to allow ETL of large datasets, saving a considerable amount of time for the data services team on ingestion.

NASA GHRC DAAC, ITSC UAH

Huntsville, AL, USA

Graduate Research Assistant

Aug 2022 - Dec 2023

- Implemented a cloud-based data subset tool using **RESTful API** services in **Python** and **dockerized serverless microservices** in **AWS**, resulting in a **70% increased efficiency** for subsetting on large Geospatial data.
- Released an **open-source Python package** (pypi.org/project/fcx-playground/) for cloud-native interactive visualization playground for NASA GHRC, providing a more comprehensive and lucid view of atmospheric science data.
- Led the development of a **cloud-native 3D Geo-visualization** tool in **ReactJS**, coupled with multiple **data processing pipelines** in **Python** and **AWS Lambdas**, hence representing large heterogeneous instrument data from NASA field Campaigns.

GURZU INC

Boston, MA, USA

Software Engineer

Aug 2020 - Aug 2022

- Designed and led the implementation of SaaS-based digital advertising software using **MERN** stack, supporting digital banner ads across urban areas, scaling to serve 200k+ viewership per day. (splashnode.com/)
- Developed a **modularized** test-sample collection **ERB template** in **Rails**, driving the swift creation of a COVID test management system during the COVID outbreak, ensuring crucial business continuity during the pandemic's peak.
- Migrated an existing **x64** electron desktop application to **arm architecture**, enabling cheaper compute devices and **80% reduced cost**.

JYAASA TECHNOLOGIES

Kathmandu, Nepal

Associate Software Engineer

Jan 2019 - Aug 2020

- Designed and developed **RESTful APIs**, and integrated them with distributed services using **RabbitMQ** as **message-passing middleware**, to implement a resilient cloud-native digital billboard solution.
- Implemented **authentication and authorization** backend using **JWT** and **Passport.js** to ensure secure and reliable API access.
- Developed an **electron** desktop app with **responsive** widgets (clock, weather, and RSS feeds) using **HTML5, CSS3, and jQuery**.

SKILLS

- Language:** *JavaScript, Python, Typescript, Java, C, C++, MATLAB, Ruby*
- Web Technologies:** *ReactJS, Redux, NodeJS, ExpressJS, Django, FastAPI, WebSockets, JWT, HTML5, CSS3, SAAS, RESTful APIs, GraphQL, MUI, Bootstrap, NPM, Yarn*
- Cloud Services Platform:** *AWS Lambda, AWS EC2, AWS Fargate, AWS S3, AWS API Gateway, AWS Cloudwatch*
- Database Technologies:** *MySQL, PostgreSQL, SQLite, MongoDB, DynamoDB*
- Data and visualization Tools:** *Numpy, Pandas, Matplotlib, Seaborn, Xarray, Cesium, CZML, 3DTiles, D3.js, Chart.js*
- Build and Other Tools:** *Terraform, Docker, Webpack, Git, Linux/Bash, Electron.js, RabbitMQ, JIRA, Confluence, Mocha, Chai*

EDUCATION

University Of Alabama In Huntsville

Masters in Computer and Information Science (GPA 3.9)

Huntsville, AL, USA

Graduation Date: Dec 2023